

NYMEX RBOB GASOLINE FUTURES (RB)

Spread & Curve Structure — Fundamental Reference

Refinery Economics, Seasonal Demand, Specifications, and the Crack Spread

This document presents a fact-based reference on the fundamental drivers of NYMEX RBOB Gasoline futures (ticker: RB), with particular focus on the forces that shape the forward curve and the spread structure between contract months. RBOB (Reformulated Blendstock for Oxygenate Blending) is the primary US gasoline benchmark and the most liquid petroleum product futures contract in North America. Its price is determined by the interaction of refinery economics, seasonal specification changes, crude oil input costs, domestic demand patterns, inventory levels, and trade flows. Unlike crude oil, RBOB is a finished product blendstock subject to regulatory specifications that change by season and region, creating recurring, predictable spread structures. Understanding the refining chain, the specification calendar, and the EIA weekly inventory cycle is the foundation of spread and butterfly trading on the RBOB curve.

Futures First — Internal Education Material
For Analyst Group Use

1. Contract Specifications and What RBOB Is

RBOB is not finished gasoline — it is the blendstock into which ethanol is blended at the terminal to produce finished retail gasoline. This distinction matters for understanding delivery economics and the relationship between RBOB futures prices and the retail pump price.

Parameter	Detail
Exchange	CME Group — NYMEX
Ticker	RB (front month)
Full name	Reformulated Blendstock for Oxygenate Blending (RBOB)
Contract size	42,000 US gallons (1,000 barrels)
Quotation	USD per gallon; minimum tick = \$0.0001/gallon = \$4.20 per contract
Contract months	All 12 calendar months; active contracts extend approximately 3 years forward. Liquid trading concentrated in front 6–12 months.
Settlement type	Physical delivery — pipeline or barge delivery into New York Harbor (NYH) area at Buckeye Pipe Line, Laurel Pipe Line, or other designated facilities
Delivery location	New York Harbor (NYH) — the primary US Atlantic Coast petroleum product hub. Includes terminals in New Jersey, Connecticut, and New York Boroughs areas.
Delivery grade	RBOB meeting ASTM D4814 specification for the applicable season (summer RVP ≤ 9.0 psi in most RFG areas; winter RVP higher). Must be pipeline-quality, non-oxygenated blendstock.
Last trading day	Last business day of the month preceding the delivery month
Trading hours	CME Globex: Sunday–Friday 18:00–17:00 ET (23-hour); liquid hours approximately 07:00–14:30 ET
Contract value at \$3/gal	\$126,000 per contract
DV01	\$0.0001/gal move = \$4.20 per contract

1.1 What Is RBOB and Why Does It Matter?

The transition from conventional gasoline to RBOB as the futures benchmark occurred in 2006 when EPA Reformulated Gasoline (RFG) programme requirements and the phase-out of MTBE (methyl tert-butyl ether) as an oxygenate made ethanol blending the standard for compliant gasoline:

- **RBOB (Reformulated Blendstock for Oxygenate Blending):** a sub-octane blendstock produced at the refinery, designed to be blended with 10% ethanol (E10) at the terminal to produce finished regular-grade gasoline (87 AKI octane). The 10% ethanol addition provides approximately 2.5–3 octane points, bringing the blend to the required 87 AKI.
- **CBOB (Conventional Blendstock for Oxygenate Blending):** the equivalent blendstock for non-RFG areas. CBOB is the product used in areas not subject to EPA RFG requirements (roughly 30% of US gasoline demand). CBOB trades at a discount to RBOB because it has less stringent vapour pressure and benzene specifications.
- **The ethanol "embedded subsidy":** because ethanol is blended into RBOB at 10% by volume, the effective crude-to-gasoline yield calculation includes the ethanol component. Changes in ethanol prices affect the economics of blending and can influence the RBOB-crude crack spread at the margin.
- **New York Harbor (NYH) as the delivery point:** NYH is the largest petroleum product import/export hub on the US East Coast. The NYH price reflects not only domestic refining economics

but also the landed cost of imported gasoline from Europe, the Caribbean, and the Middle East. This import arbitrage is a key variable for RBOB prompt spreads.

1.2 The RBOB-WTI Crack Spread

The single most important derived price relationship in RBOB trading is the crack spread — the refinery margin for producing RBOB from crude oil:

- **The simple RBOB crack:** RBOB price (\$/gallon × 42 gallons/barrel) minus WTI crude price (\$/barrel). This represents the gross refining margin per barrel of crude processed into gasoline (before costs and other products).
- **The 3-2-1 crack spread:** $(2 \times \text{RBOB } \$/\text{bbl} + 1 \times \text{ULSD } \$/\text{bbl} - 3 \times \text{WTI } \$/\text{bbl}) \div 3$. A simplified refinery margin assuming 2 barrels of gasoline and 1 barrel of diesel from 3 barrels of crude. Wide crack spreads incentivise refineries to maximise runs; narrow or negative cracks incentivise run cuts.
- **Crack spread seasonality:** RBOB crack spreads follow a highly seasonal pattern. They typically widen from February through May (spring refinery maintenance reduces product supply while driving season demand builds) and narrow from June through October (maintenance ends, production increases, driving season demand peaks and then falls). This seasonal pattern is one of the most documentable recurring patterns in petroleum markets.

2. Gasoline Specifications and the Seasonal Specification Calendar

Gasoline product specifications change by season and by region. These regulatory changes create predictable, calendar-driven shifts in the RBOB forward curve that are unique to gasoline among commodity futures markets. Understanding the specification calendar is essential for interpreting spread behaviour around the transition dates.

2.1 Reid Vapour Pressure (RVP) — The Core Specification

Reid Vapour Pressure (RVP) measures the volatility of gasoline (how easily it evaporates at a standard temperature). Higher RVP means more volatile — problematic in summer because it contributes to ground-level ozone (smog) formation when gasoline evaporates from vehicle fuel systems and storage tanks. EPA regulations mandate lower RVP in summer to reduce smog:

- **Summer-grade gasoline:** $RVP \leq 9.0$ psi for conventional gasoline in most markets. For EPA Reformulated Gasoline (RFG) areas (major urban markets — New York, Los Angeles, Chicago, Philadelphia, Houston, and others): $RVP \leq 7.8$ psi (9 psi specification minus 1 psi waiver for E10 ethanol blending). Summer-grade is more expensive to produce: refineries must remove lighter, higher-RVP components (primarily butane and pentane), which reduces gasoline yield and requires more sophisticated refinery configuration.
- **Winter-grade gasoline:** higher RVP allowed (up to 15 psi in some regions). Refiners can blend more butane into gasoline (butane is cheap, increases volume, and raises RVP). This "butane blending" economics significantly affects the winter-grade vs summer-grade price differential.
- **The specification transition dates:** summer-grade must be at terminals and retail stations by June 1 in most RFG areas (and May 1 in California). To ensure summer-grade is available at the pump by June 1, refineries typically begin producing summer-grade by March 1–15, and terminals begin transitioning from February 15 onward. The February RBOB contract is the last contract that can be winter-grade. The April contract must be summer-grade.

2.2 The February-March "Grade Change" Spread

The transition from winter-grade to summer-grade creates one of the most structurally reliable spread moves in RBOB:

- Summer-grade RBOB is more expensive to produce than winter-grade because: (1) the removal of high-RVP butane reduces refinery blending economics, (2) the refinery must produce a more tightly specified product, and (3) the first summer-grade production comes on amid refinery maintenance season (which reduces total product output).
- The March RBOB contract must price summer-grade product; the February contract prices winter-grade. The Feb-March spread therefore contains a structural grade-change premium embedded in the March price, which is distinct from a standard month-to-month carry spread.
- Historically, the March-April spread and the broader February-April spread widen from late autumn into spring as the market builds in the grade-change and maintenance premium. The magnitude varies with the state of refinery maintenance and the tightness of product inventories heading into spring.

2.3 Boutique Fuels and California Specifications

- **California Air Resources Board (CARB) gasoline:** California operates under its own, stricter gasoline specification (CARB Phase 3 reformulated gasoline) that differs from federal RFG requirements. CARB gasoline requires specific benzene, aromatic, and sulphur limits stricter than federal standards. Only a limited number of refineries — primarily those in California — can economically produce CARB gasoline. This creates a semi-isolated California market where CARBOB (CARB blendstock) trades at a premium to RBOB and can spike dramatically when California refinery

outages occur.

- **Regional boutique fuel programmes:** beyond California, several other states and urban areas have adopted unique gasoline specifications (Arizona summer, Dallas-Fort Worth summer, etc.) that create local pricing pockets and limit the interchangeability of gasoline across regions. The proliferation of boutique fuels reduces the flexibility of the physical gasoline distribution system to respond to regional supply disruptions.
- **E15 and higher ethanol blends:** EPA has approved E15 (15% ethanol) for all model year 2001+ vehicles. Seasonal E15 sale restrictions (previously banned in summer due to RVP waiver limitations) have been partially relaxed. Wider E15 adoption would change the blendstock economics for RBOB over time (lower RBOB requirement per gallon of pump gasoline).

3. Supply — Refinery Economics and Production

RBOB supply is entirely a function of refinery operations. Unlike crude oil, gasoline cannot be produced from a wellhead. Understanding the refinery system — its geography, complexity, maintenance cycles, and capacity — is the core of RBOB supply analysis.

3.1 US Refinery Landscape

PADD	Region	Refining Capacity	Gasoline Role	RBOB Curve Impact
PADD 1	East Coast	~1.1 mb/d	Minimal domestic refining. Highly dependent on imports from Europe, Caribbean, and pipeline from Gulf Coast. Primary demand centre for NYH RBOB.	Import economics (ARA vs NYH arb) directly affect prompt RBOB supply. PADD 1 inventory levels most relevant for NYH delivery.
PADD 2	Midwest	~4.0 mb/d	Significant refining capacity. Primarily serves domestic Midwest demand. Exports excess to East Coast via Colonial/Plantation pipelines.	PADD 2 gasoline stocks and utilisation affect Colonial Pipeline throughput, which supplies PADD 1 and indirectly influences NYH RBOB pricing.
PADD 3	Gulf Coast	~10.3 mb/d	Largest US refining centre. Major producer of RBOB for both Gulf Coast consumption and export to PADD 1 via Colonial/Plantation. Also significant exporter of gasoline internationally.	Gulf Coast refinery utilisation, maintenance schedules, and hurricane impacts are the primary domestic supply variables for the RBOB curve.
PADD 4	Rocky Mountain	~0.6 mb/d	Small, isolated regional refiners. Serves local market.	Minimal direct impact on NYH RBOB pricing.
PADD 5	West Coast/AK	~2.9 mb/d	Significant capacity but isolated from East Coast. Produces CARB gasoline. Subject to California refinery risk premium.	CARB refinery outages can spike CARBOB prices and draw national attention but rarely affect NYH RBOB directly.

3.2 Refinery Maintenance — The Spring Turnaround Season

Planned refinery maintenance is the most predictable supply-side driver of the RBOB forward curve. The spring turnaround season is the dominant annual maintenance period:

- **Spring turnaround season (February-April):** refineries conduct planned maintenance on catalytic crackers (FCC units), hydrotreaters, reformers, and other units during this period for several reasons: (1) winter demand has passed and summer demand has not yet peaked, creating a lower-impact maintenance window; (2) it allows refineries to prepare for the transition to summer-grade gasoline production. Turnarounds reduce crude runs by 5-15% at affected refineries for 3-8 week periods.
- **Effect on RBOB spreads:** the spring turnaround season reduces gasoline production simultaneously with rising driving season demand, creating the most consistent annual tightening of the February-May RBOB spread. This spread typically widens from late January onward as the market prices in the combined maintenance and specification-change premium.

- **Autumn turnaround season (September-October):** a smaller secondary maintenance window. Refineries transition from summer-grade to winter-grade production and conduct shorter maintenance runs. The autumn turnaround creates a more modest seasonal tightening in the September-October spread, which competes with the seasonal demand decline post-summer.
- **Fluid Catalytic Cracking (FCC) unit maintenance:** the FCC is the most gasoline-intensive refinery unit — it converts heavier crude fractions into lighter products, primarily gasoline. An FCC outage at a major refinery (e.g. Motiva Port Arthur, Marathon Galveston Bay, LyondellBasell Houston) can remove 50,000–100,000 b/d of gasoline production, which is an immediately market-moving event for the prompt RBOB contract.

3.3 Gulf Coast Refinery Geography and Hurricane Risk

- Approximately 50% of US refining capacity is located on the Gulf Coast (Texas and Louisiana). This geographic concentration creates a structural vulnerability to Atlantic hurricane disruptions:
- A major hurricane making landfall on the Texas or Louisiana coast can shut in 1.0–3.0 mb/d of refinery capacity for 1–4 weeks (operational shutdown, storm preparation, flooding risk). Post-storm restarts typically take 1–2 weeks for unaffected plants and months for flooded or damaged facilities.
- Hurricane Katrina (2005) and Hurricane Harvey (2017) are the two most significant documented refinery disruption events. Harvey shut in approximately 4.4 mb/d of Gulf Coast refinery capacity (including Motiva Port Arthur — the largest US refinery), creating a simultaneous crude oversupply (refineries not running → not buying crude) and product shortage (no gasoline production). RBOB spot prices rallied sharply while WTI weakened — RBOB crack spreads expanded dramatically.
- Hurricane season (June 1 – November 30) adds an annual risk premium to summer and early autumn RBOB contracts, concentrated in Gulf Coast-exposed months (July, August, September, October).

3.4 Refinery Complexity and the Yield Curve

Not all refineries produce the same proportion of gasoline from a barrel of crude. Complexity determines yield:

- **Simple refineries (topping units / hydroskimming):** process crude into straight-run products. Lower gasoline yield (~30–35% of crude barrel). Cannot process heavy crude economically.
- **Complex refineries (cracking, coking, hydrocracking):** use FCC, hydrocracking, and/or delayed coking units to convert heavier fractions into lighter products. Higher gasoline yield (~45–50% of crude barrel). More profitable when light-heavy crude differentials are wide.
- **The Nelson Complexity Index:** measures refinery complexity. Higher complexity = higher capital cost but greater flexibility to process different crude grades and maximise higher-value product yields. Gulf Coast refineries have among the highest complexity ratings globally.
- **Light/heavy crude differential and gasoline yield:** complex refineries designed to process heavy crude tend to produce more gasoline and diesel per barrel when heavy crude is cheap relative to light crude. The WTI-WCS (Canadian heavy) or WTI-Mars differential affects the gasoline production economics of complex Gulf Coast refineries.

4. The Colonial Pipeline System — Gasoline's Supply Artery

The Colonial Pipeline is the most important piece of petroleum product infrastructure in the United States for RBOB pricing. It is the primary supply mechanism for connecting Gulf Coast refinery production to the PADD 1 (East Coast) demand market, which is the delivery region for NYMEX RBOB futures.

4.1 Colonial Pipeline Infrastructure

- **Route:** runs from Houston, Texas to Linden, New Jersey — approximately 5,500 miles. Passes through Alabama, Georgia, South Carolina, North Carolina, Virginia, Maryland, and New Jersey/Pennsylvania.
- **Capacity:** approximately 2.4 million barrels per day total across two main lines — Line 1 (gasoline and aviation fuel) and Line 2 (diesel, heating oil, jet fuel). RBOB and gasoline represent the majority of Line 1 throughput.
- **Transit time:** approximately 18–20 days from Houston to New York. This means that gasoline produced at a Gulf Coast refinery today will arrive at the NYH terminal area in approximately 3 weeks. This transit time creates a physical market lag between Gulf Coast production changes and NYH inventory responses.
- **Supplementary pipeline: Plantation Pipe Line:** a second major pipeline from the Gulf Coast to the Southeast (Augusta, Georgia terminus). Less direct impact on NYH pricing but serves Southeast demand markets.

4.2 Colonial Pipeline Disruptions

- **September 2016 Alabama spill:** Colonial Line 1 ruptured near Shelby County, Alabama, removing approximately 900,000 b/d of gasoline throughput for approximately 12 days. PADD 1 gasoline inventories fell sharply; RBOB spot prices at NYH spiked \$0.25–0.35/gallon above Gulf Coast prices, reflecting the regional dislocation.
- **May 2021 ransomware attack (DarkSide):** Colonial Pipeline shut down all operations for approximately 6 days following a ransomware cyberattack. PADD 1 gasoline shortages emerged within 3–4 days at the retail level (particularly in the Southeast). RBOB prompt prices spiked and the prompt spread widened as the market priced near-term supply disruption.
- **Structural implications for spread traders:** Colonial Pipeline scheduling and operational status is a constant background variable for RBOB spread traders. Any reduction in Colonial throughput disproportionately affects PADD 1 supply and the NYH delivery market — the benchmark for NYMEX RBOB.
- **Space availability and scheduling:** Colonial Pipeline operates on a cyclical batch scheduling system. Shippers must nominate volumes by specific deadlines. Tight cycles (high demand relative to capacity) can delay the arrival of barrels at NYH, creating temporary prompt tightness even without a physical disruption.

4.3 Import Arbitrage — The ARA-NYH Route

The East Coast is structurally short of domestic refining capacity and imports significant volumes of gasoline from Europe (primarily the Amsterdam-Rotterdam-Antwerp, or ARA, hub) and the Caribbean. This import arbitrage is a crucial supply variable for RBOB prompt pricing:

- **The ARA-NYH arbitrage:** when NYH RBOB prices are sufficiently above ARA gasoline prices (accounting for shipping costs of approximately \$0.03–0.08/gallon and transit time of approximately 7–10 days by tanker), European and Atlantic Basin gasoline flows toward New York. This arbitrage is a natural price ceiling for NYH RBOB: if NYH rises far enough above ARA, imports flow and supply the market.

- **European gasoline specification compatibility:** European gasoline standards (EN 228) differ from US RFG requirements. European gasoline can be sold as CBOB in non-RFG areas but requires adjustment (specifically regarding benzene content and aromatics specifications) for use in RFG areas as RBOB. This limits the volume that can be directly substituted and means the ARA-NYH arb is not perfectly elastic.
- **Caribbean refinery capacity:** refineries in the US Virgin Islands (Limetree Bay), Venezuela (Paraguana complex — historically), and other Caribbean locations have at times supplied gasoline to PADD 1. Venezuelan refinery capacity constraints reduced this supply source after 2018.

5. Demand Fundamentals

5.1 US Gasoline Demand — The Overall Picture

US gasoline demand has been the subject of significant structural debate. The following facts characterise the market:

- Total US gasoline demand runs at approximately 8.5–9.0 million barrels per day (357–378 million gallons per day), making the US the world's largest gasoline market by volume.
- US gasoline demand peaked in 2007 at approximately 9.3 mb/d and declined through the financial crisis. It recovered partially but has not returned to the 2007 peak. The combination of improved vehicle fuel economy (CAFE standards requiring higher MPG), slower vehicle miles travelled (VMT) growth, and growing electric vehicle (EV) adoption has kept demand below the pre-crisis peak.
- EV adoption reduces gasoline demand per vehicle replaced, but the pace of fleet turnover is measured in years (the average US vehicle age is approximately 12–13 years). Near-term structural gasoline demand decline from EV adoption is a slow-moving phenomenon for the 1–5 year forward curve.

5.2 The Driving Season — The Most Important Demand Cycle

Gasoline demand follows a pronounced annual seasonal pattern driven by the US driving season:

- **Memorial Day (late May) to Labor Day (early September):** the peak US driving season. Americans drive significantly more during summer holidays, road trips, and vacation travel. Weekly EIA gasoline demand (implied by product supplied) typically runs 5–10% above the off-season level during this period. This seasonal demand increase is the primary driver of the RBOB summer premium.
- **The Memorial Day weekend "demand signal":** Memorial Day weekend gasoline demand is closely watched as the opening indicator of summer driving season strength. A strong Memorial Day implies a bullish summer; weak demand that weekend can signal a softening of the seasonal premium.
- **July 4th and Labor Day:** the two other major holiday demand events within the driving season. These represent the peak weeks of annual gasoline demand and can temporarily support the July and August RBOB contracts relative to adjacent months.
- **Post-Labor Day demand drop:** US gasoline demand falls sharply after Labor Day as school starts, vacation travel ends, and commuting patterns normalise. This creates a predictable seasonal weakening of the September and October RBOB contracts relative to the August peak.

5.3 Vehicle Miles Travelled (VMT) and Gasoline Consumption

- **VMT data (FHWA):** the Federal Highway Administration publishes monthly traffic volume data (VMT) with approximately a 2-month lag. VMT is a physical demand indicator — higher VMT = more gasoline consumed. Year-over-year VMT changes provide a structural demand signal.
- **Fuel efficiency (MPG):** as average fleet fuel efficiency improves (CAFE standards have increased average new vehicle MPG from ~25 mpg in 2010 to ~30+ mpg for new vehicles in 2023), the same VMT requires less gasoline. The effective MPG of the entire in-use fleet (not just new vehicles) determines the VMT-to-gasoline conversion ratio.
- **Retail gasoline price elasticity:** short-run price elasticity of gasoline demand is low (approximately -0.05 to -0.10) — a 10% rise in pump price reduces demand by 0.5–1.0% in the short run. Long-run elasticity is higher (approximately -0.20 to -0.30) as consumers adjust vehicle choices and commuting patterns. High pump prices during the spring demand build can modestly suppress the seasonal demand ramp.

5.4 Export Demand — The Growing Variable

- **US gasoline exports:** approximately 800,000–1,200,000 b/d. The US has become a significant net exporter of gasoline, primarily to Mexico and Latin America. Export demand provides a floor for RBOB prices: when domestic supply is abundant and RBOB is cheap, export opportunities absorb excess product.
- **Mexico as the primary export destination:** Mexico imports approximately 400,000–600,000 b/d of gasoline from the US, primarily via Gulf Coast pipelines and tankers. Mexican gasoline import demand is relatively stable, driven by the gap between domestic Pemex refinery output and domestic consumption.
- **Latin American export demand:** Central America, Colombia, and other Latin American markets import US gasoline when the arbitrage is open (US export price plus shipping below local market price). This demand is more price-sensitive than Mexico's structural import need.

6. EIA Weekly Data — The Primary Scheduled Signal

The EIA Weekly Petroleum Status Report (published every Wednesday at 10:30 ET) is the dominant scheduled weekly data event for RBOB spreads. The gasoline-specific components are:

Data Point	What It Measures	Signal Interpretation	Curve Impact
Total US gasoline stocks (mb)	All gasoline (finished + blendstock + ethanol) in primary storage. Broken out by PADD.	Above 5-yr avg = bearish supply signal; below = bullish. Year-over-year deviation matters more than absolute level.	Immediate C1-C2 repricing. PADD 1 and PADD 2 stocks most relevant for NYH RBOB.
Gasoline production (mb/d)	Domestic refinery gasoline output for the week.	High production = bearish for near-term prices; low = bullish. Most relevant during refinery maintenance season.	C1-C3; most impactful when combined with inventory signal
Gasoline imports (mb/d)	Volume of finished gasoline and blendstock imported (primarily to PADD 1).	High imports signal arbitrage from ARA is open — modestly bearish for RBOB premium. Low imports signal arb is closed and domestic market must be supplied internally.	C1-C2; PADD 1 specific but transmitted to NYH RBOB pricing
Gasoline exports (mb/d)	Volume of gasoline exported (primarily from PADD 3 via Gulf Coast).	High exports tighten domestic supply; bullish for near-term RBOB.	C1-C3; affects total supply/demand balance
Gasoline product supplied (mb/d)	Implied demand — calculated as production + imports — exports — stock change. The EIA's weekly demand proxy.	Above-seasonal implies demand is strong; below implies weak. This is the most important demand variable in the weekly report.	C1-C4; key for summer spread direction. Memorial Day week is most watched.
Refinery crude runs and utilisation (by PADD)	How much crude refineries are processing (total and by region). Utilisation rate as % of capacity.	High utilisation = high product output (bearish for product price, bullish for crude demand). Low utilisation = maintenance or margin-driven run cuts (bullish for product price).	C1-C3; PADD 3 utilisation most relevant for RBOB production volume

6.1 PADD 1 Gasoline Inventory — The Most Relevant Regional Stock

- PADD 1 (East Coast) gasoline inventories are the most directly relevant for NYMEX RBOB futures because the NYH delivery point is in PADD 1. When PADD 1 gasoline stocks are low (below the 5-year average range), the premium of NYH RBOB over Gulf Coast (Group 3) gasoline widens, incentivising more Colonial Pipeline nominations and imports.
- PADD 1 gasoline stocks exhibit a strong seasonal pattern: they typically build during Q4-Q1 (low driving demand, refinery production continues) and draw during Q2-Q3 (driving season demand outpaces supply). The trough typically occurs in August-September.
- When PADD 1 stocks enter the driving season (Memorial Day) below the 5-year average, the seasonal supply tightening is amplified and the summer RBOB spread widens more than usual.

6.2 The Blending Stock Distinction

- The EIA reports gasoline stocks in three categories: finished gasoline (ready for retail), blending components (including RBOB, CBOB, and alkylates), and fuel ethanol (held separately). NYMEX RBOB futures settle against blending components at NYH, not finished gasoline. The composition of the

stock can matter: a build in finished gasoline is more bearish than a build in blending components, which still requires ethanol addition before retail delivery.

7. The Forward Curve Structure — Zones and Seasonal Patterns

RBOB has the most clearly seasonal forward curve of any petroleum futures contract. The combination of the summer/winter specification change and the driving season demand cycle creates a recurring annual curve shape that is more predictable than crude oil or natural gas.

Zone / Contract	Key Structural Feature	Primary Driver	Typical Spread Behaviour
Jan-Feb (winter grade)	Winter-grade specification. Butane blending economics. Low driving demand.	Refinery utilisation, import economics, PADD 1 stock rebuild post-autumn	Mild contango typical as market builds inventory ahead of grade-change. Jan-Feb spread reflects winter carry economics.
Feb-Apr (grade change)	The most structurally significant spread in RBOB. Feb = last winter-grade contract. Mar/Apr = first summer-grade contracts. Spring turnaround season reduces refinery output.	Refinery maintenance schedule, specification change premium, PADD 1 inventory entering spring	Feb-March and March-April spreads typically widen from November onward, peaking around February-March. The widest and most reliable seasonal spread in RBOB.
Apr-May (spring build)	Driving season approaching. Summer-grade production ramping. Refineries exiting turnaround.	Refinery restart pace, import arb status, PADD 1 stock trajectory	Spreads can be volatile — sometimes mildly contango if stocks rebuild fast; backwardation if stocks enter driving season low.
May-Aug (driving season)	Peak demand period. Summer specification in effect. Holiday demand spikes (Memorial Day, July 4, Labor Day).	Weekly EIA demand data (product supplied), VMT, hurricane risk, PADD 1 stocks vs 5-yr avg	Front contracts can be in backwardation if demand is outpacing supply. Summer contracts trade at peak annual premium relative to autumn.
Sep-Oct (shoulder / autumn)	Driving season ends post-Labor Day. Demand falls. Refineries begin transitioning to winter-grade. Autumn turnaround season.	Post-summer demand decline pace, autumn maintenance schedule, export demand	Prompt spreads typically weaken (demand falling). The Sep-Oct spread can be mildly contango as stocks rebuild for winter.
Nov-Dec (winter approach)	Winter-grade product. Low driving demand. Refinery configuration optimised for winter products (heating oil / diesel competes with gasoline for crude runs).	Heating oil vs gasoline yield preference (crack spread relative economics), import economics	Typically quiet, mild contango. The Nov-Dec spread is less watched than the spring spread. Mid-distillate competition for crude run allocation relevant.

7.1 The RBOB-Crude Crack Spread Seasonal Pattern

The RBOB crack spread (RBOB in \$/barrel minus WTI in \$/barrel) follows one of the most documented seasonal patterns in commodity markets:

- **January-February:** crack spreads typically at or near seasonal lows. Refinery runs are at a seasonal high from the autumn maintenance period, winter-grade production is fully underway, and driving demand is at its annual trough.
- **February-May:** crack spreads widen seasonally. Spring turnaround reduces gasoline output; specification change adds production cost; driving season demand begins building. This is the most consistent annual widening of the crack spread.
- **May-June:** crack spreads typically at or near seasonal peak. Refinery maintenance concludes, runs ramp, and the crack spread begins to normalise as production catches up with demand.

- **July-September:** crack spreads moderate from the spring peak as full summer refinery runs supply the market. Hurricanes can cause temporary spikes (Harvey 2017: RBOB crack briefly exceeded \$40/bbl).
- **October-December:** crack spreads at annual low territory again as driving season ends and winter-grade production lowers the specification cost. Heating oil competition for crude run allocation can periodically support the gasoline crack if cold weather drives heating oil demand.

The seasonal crack spread pattern is robust across years but varies in magnitude. In years of low crude oil supply or high demand, the entire crack spread curve shifts higher; in years of high refinery utilisation and abundant crude, it shifts lower. The seasonal SHAPE is more reliable than the LEVEL.

8. Regulatory and Policy Drivers

8.1 The Renewable Fuel Standard (RFS) and RINs

The Renewable Fuel Standard (RFS) mandates the blending of biofuels (primarily ethanol) into the US gasoline supply. Compliance is tracked via Renewable Identification Numbers (RINs), and the RIN market directly affects RBOB economics:

- **D6 RINs (conventional ethanol RINs):** generated when ethanol is blended into gasoline. Obligated parties (refiners, importers) must retire RINs to demonstrate compliance with the annual Renewable Volume Obligation (RVO). The price of D6 RINs represents the compliance cost and is embedded in the economics of RBOB vs finished gasoline.
- **RIN price spikes:** when RIN prices rise sharply (as occurred in 2013–2014 and 2021–2022), they increase the effective cost of gasoline production for refiners who are net short RINs (primarily merchant refiners like PBF Energy, Par Pacific). RIN price spikes can widen the spread between RBOB and finished gasoline, and disproportionately affect smaller, less-integrated refiners.
- **Small Refinery Exemptions (SREs):** the EPA can grant exemptions to small refineries from RVO compliance. Court decisions and EPA policy changes on SRE approvals directly affect aggregate RIN demand and prices, which feed through to RBOB spread economics.
- **E15 and higher blend waivers:** EPA decisions to allow E15 year-round or to expand E15 availability can affect RBOB blending economics (more ethanol means less RBOB required per gallon of pump gasoline).

8.2 EPA Tier 3 Sulphur Standards

- **Tier 3 gasoline sulphur standard:** EPA Tier 3 regulations (phased in from 2017) reduced the maximum sulphur content of US gasoline from 30 ppm (Tier 2) to 10 ppm (Tier 3 average). This required significant refinery investment in desulphurisation (hydrotreating) capacity.
- Tier 3 compliance costs are now embedded in refinery economics and are not a current market-moving variable but represent a structural cost that defines the minimum RBOB-crude crack spread at which new refinery investment is economically justified.

8.3 Strategic Petroleum Reserve (SPR) Product Releases

- The US SPR holds crude oil, not refined products. However, in extraordinary circumstances, the Department of Energy has used the Northeast Home Heating Oil Reserve and other product stockpiles to address regional shortages. More commonly, SPR crude releases indirectly affect RBOB by increasing crude supply available to Gulf Coast refineries, which can increase refinery runs and gasoline production.
- After Hurricane Katrina (2005), the IEA coordinated release of 60 million barrels of crude specifically helped supply Gulf Coast refineries as they restarted — an example of SPR impacting the product supply chain rather than crude prices directly.

9. Cross-Market Relationships

9.1 RBOB and WTI — The Crack Spread Relationship

The RBOB-WTI crack spread is the fundamental economic relationship that connects the crude oil input cost to the gasoline output price. It is simultaneously a production economics indicator and a spread trading instrument:

- **Crack spread = refinery margin proxy:** wide crack spreads incentivise maximum refinery throughput (buying crude, producing gasoline) — bullish for crude demand, bearish for gasoline going forward (higher production). Narrow crack spreads incentivise run cuts — bearish for crude demand, bullish for gasoline going forward.
- **Light-heavy crude differential and RBOB crack:** complex refineries designed to process heavy crude earn a premium crack spread when heavy crude is cheap. Widening WTI-WCS or Brent-Dubai (sour) differentials (heavy crude cheapening relative to light) benefit complex Gulf Coast refiners and can increase their profitability and run rates — moderately bearish for RBOB on the supply side.
- **RBOB vs Brent crack:** European refiners who supply PADD 1 gasoline imports are more sensitive to the RBOB-Brent crack (since their crude input is priced off Brent). When RBOB-Brent crack is wide, European refinery economics are attractive and imports flow to NYH. When narrow, imports slow and domestic supply must fill the gap.

9.2 RBOB and ULSD — The Product Crack Spread Pair

- **ULSD (Ultra-Low Sulphur Diesel) NYMEX futures (ticker: HO):** the sister contract to RBOB, also priced in \$/gallon at NYH. RBOB and ULSD are co-produced in the refinery and compete for the same crude input. When diesel demand is strong and diesel cracks are wide, refineries may maximise diesel yield at the expense of gasoline — supportive for RBOB (less supply) while ULSD cracks compress.
- **The RBOB-ULSD spread:** normally RBOB trades at a premium to ULSD on a \$/gallon basis in summer (driving season demand) and ULSD trades at a premium in winter (heating oil / distillate demand). Divergences from this seasonal norm — e.g. unusually high diesel demand due to cold weather or high heating oil demand — can affect refinery yield decisions and feed through to RBOB supply.
- **The 3-2-1 crack spread:** $(2 \times \text{RBOB} + 1 \times \text{ULSD} - 3 \times \text{WTI}) / 3$. This combined measure of refinery margin is the most widely watched proxy for overall refining profitability. Monitoring the seasonal pattern of the 3-2-1 crack provides context for whether spread moves are driven by RBOB-specific supply/demand or broader refinery economics.

9.3 Naphtha and International Gasoline Blending

- **Naphtha:** a lighter crude fraction used both as a petrochemical feedstock (steam cracking for ethylene) and as a gasoline blending component. When naphtha is cheap relative to RBOB, blenders can substitute naphtha into the gasoline pool, reducing RBOB demand at the margin. The naphtha-RBOB spread is relevant for understanding blending economics in the international market.
- **ARA gasoline stocks:** the Amsterdam-Rotterdam-Antwerp gasoline inventory (reported weekly by Insights Global / Genscape) determines how much European product is available for export to PADD 1. Rising ARA stocks when the ARA-NYH arb is open is a bearish signal for RBOB prompt spreads.
- **Singapore gasoline market:** Asia-Pacific gasoline demand and refinery economics affect global gasoline trade flows, though the direct impact on RBOB is secondary compared to the ARA-NYH arb.

9.4 Ethanol Prices and RBOB Blending Economics

- **Chicago ethanol price:** CBOT ethanol futures (ticker: EH) price corn-based ethanol at the Chicago delivery point. Ethanol is mandated at 10% volume in virtually all US gasoline. When ethanol trades at a significant discount to RBOB (on an energy-equivalent basis), blending ethanol into RBOB creates a positive blending margin (the "splash blend" economics). This blending profit is captured by terminal operators and blenders, not refiners — but it affects the effective supply of finished gasoline.
- **Corn prices and ethanol supply:** corn is the primary feedstock for US fuel ethanol. When corn prices are high, ethanol production economics deteriorate and ethanol supply can tighten at the margin, which can affect the economics of blending and indirectly the RBOB-ethanol differential.

10. Documented Seasonal Patterns

RBOB gasoline has the most reliable and well-documented seasonal patterns of the major petroleum futures contracts. The combination of specification changes, driving season demand, and refinery maintenance creates a recurring annual structure that is the foundation of spread positioning.

Period	Specification	Supply Factor	Demand Factor	Typical Spread Direction	Reliability
Nov-Jan (pre-spring build)	Winter grade	Refinery runs high; ample production; imports light	Lowest driving demand; holiday traffic modest	Mild contango; market builds inventory ahead of spring	High
Feb-Apr (grade change + maintenance)	Transition to summer	Spring turnaround reduces output; summer spec costs more to produce	Driving season beginning to build	Front spreads widen (backwardation possible); grade change premium in March	Highest — most reliable seasonal move
May-Jun (driving season start)	Summer grade	Maintenance concluding; runs ramping; import arb may open	Memorial Day driving peak; steady summer demand building	Spreads can narrow as supply responds to demand; volatile depending on inventory vs seasonal expectation	Moderate
Jul-Aug (peak driving)	Summer grade	Full refinery utilisation; hurricane risk adds supply uncertainty	Peak seasonal demand; July 4 and summer vacation travel	Front contracts firm; backwardation possible if stocks low	Moderate — hurricane risk adds unpredictability
Sep-Oct (post-summer)	Transition to winter	Autumn turnaround; transitioning to winter-grade (cheaper to produce)	Sharp post-Labor Day demand drop	Spreads weaken; contango typically returns as demand falls	High
Oct-Nov (pre-winter)	Winter grade	Butane blending economics maximise winter-grade production	Continued weak driving demand	Seasonal low in crack spreads; RBOB discounted vs ULSD	High

10.1 Key Documented Historical Spread Events

The following events illustrate how the structural factors described in this document have manifested in historical spread moves:

- **Spring 2012 — extreme grade-change premium:** refinery maintenance coincided with low PADD 1 stocks entering the specification-change period. The March-April RBOB spread widened significantly; RBOB crack spreads reached \$35-40/bbl in April 2012, well above historical norms for that period.
- **August 2017 — Hurricane Harvey:** Harvey shut in approximately 4.4 mb/d of Gulf Coast refinery capacity. RBOB front-month spiked ~\$0.30/gallon in two days. RBOB-WTI crack spread reached ~\$35/bbl (from ~\$18 pre-storm) as product shortages emerged while crude was landlocked at Cushing (refineries not running). Colonial Pipeline also operated at reduced throughput.
- **September 2016 — Colonial Pipeline spill:** PADD 1 gasoline prices decoupled from Gulf Coast for approximately 12 days. RBOB at NYH traded \$0.25-0.35/gallon above Group 3 gasoline — the ARA import arb opened and European cargoes were chartered for immediate loading to NYH.
- **Spring 2022 — post-Ukraine supply shock:** sanctions on Russia and reduced European refinery activity reduced European gasoline export availability to PADD 1. RBOB crack spreads reached

record highs of \$50+ /bbl in June 2022 as the ARA import arb partially closed and domestic production was insufficient to rebuild stocks.

11. Macro and Financial Drivers

11.1 Crude Oil Price Level and RBOB Spread Independence

RBOB spreads (calendar spreads between months) are largely independent of the level of crude oil prices. A \$10/barrel rise in WTI shifts the entire RBOB curve higher in approximately parallel fashion, but does not, by itself, change the structure of the spreads. Spread traders are positioned for changes in the SLOPE of the curve, not the level — and the slope is driven by the supply/demand and seasonal factors described in previous sections.

- However, extreme crude oil price levels can indirectly affect RBOB spreads by: (1) at very high crude prices, driving demand can be suppressed (consumers reduce discretionary driving), which can mute the seasonal driving season premium; (2) at very low crude prices, the full-carry contango for crude becomes very wide, which can spill over into product markets and incentivise product storage.

11.2 US Dollar and Import Competitiveness

- A stronger USD makes imported gasoline (from Europe) cheaper in USD terms for PADD 1 buyers, which can increase the competitiveness of the ARA-NYH import arb and add supply to the East Coast market — modestly bearish for prompt RBOB.
- A weaker USD makes US gasoline exports cheaper in foreign currency terms, increasing export demand from Mexico and Latin America — modestly bullish for US gasoline prices.

11.3 Economic Activity and Gasoline Demand

- **GDP and consumer confidence:** gasoline demand is correlated with economic activity — higher GDP growth → more freight movement (diesel demand) and consumer spending including leisure driving (gasoline demand). Consumer confidence indices (Conference Board, University of Michigan) are leading indicators of discretionary driving and travel.
- **Unemployment rate:** lower unemployment = more commuting trips = more gasoline consumption. The labour market is a slow-moving but real driver of baseline gasoline demand.

11.4 Speculative Positioning — COT Data

- **CFTC COT for RBOB:** weekly managed money net position in RBOB futures. As with other energy contracts, extreme net long positioning (particularly heading into the driving season) signals crowding risk and an elevated probability of mean-reversion if demand disappoints.
- **Producer/merchant hedging:** refiners hedge forward RBOB production (selling futures to lock in crack margins). Heavy refiner hedging in the summer months (visible as large commercial short positions in summer RBOB contracts) reflects confidence in summer margins and creates known selling pressure.

12. Key Data Sources and Release Schedule

Report	Publisher	Frequency	Release Timing	Primary RBOB Curve Impact
EIA Weekly Petroleum Status Report — Gasoline stocks, production, imports, demand (product supplied)	US EIA	Weekly	Wed 10:30 ET	Prompt C1-C3 — the dominant weekly event; PADD 1 stocks most relevant for NYH RBOB
EIA Short-Term Energy Outlook (STEO)	US EIA	Monthly	2nd Tue of month	C2-C12 — supply/demand balance, refinery utilisation, demand forecast
Insights Global / Genscape ARA gasoline stocks	Private	Weekly	Wed-Thu	C1-C2 — European gasoline supply available for NYH import; ARA-NYH arb signal
Colonial Pipeline operational status and scheduling	Colonial / FERC filings	Ongoing	Real-time	C1-C2 — any throughput restriction immediately affects PADD 1 supply; most acute in summer
Baker Hughes Rig Count	Baker Hughes	Weekly	Fri ~13:00 ET	C3-C12 — leading indicator for crude production, which feeds refinery throughput and gasoline output
EIA Refinery Utilisation / Crude Runs (PADD 3)	US EIA	Weekly with STSR	Wed 10:30 ET	C1-C3 — Gulf Coast runs determine gasoline production volume
AAA Daily Retail Fuel Price	AAA	Daily	Morning	Consumer demand signal; media attention at high prices can affect short-run driving behaviour
FHA Vehicle Miles Travelled (VMT)	FHWA	Monthly	~2-month lag	C3-C6 — structural demand signal; confirms or revises seasonal demand thesis
CFTC Commitments of Traders	CFTC	Weekly	Fri ~15:30 ET	Positioning / crowding — all months; managed money seasonal pattern most relevant
EPA RIN prices (D6 ethanol RINs)	EPA / OPIS / Platts	Daily	Business hours	C1-C6 — compliance cost embedded in RBOB economics; spikes in RIN prices affect refiner margins
NOAA Atlantic Hurricane Forecast	NOAA NHC	Active season (Jun-Nov)	Daily updates	C3-C5 — Gulf Coast refinery disruption risk; most relevant for summer/early autumn contracts
EIA Annual Energy Outlook	US EIA	Annual	Jan-Feb	C24+ — long-run gasoline demand trajectory; EV adoption and fuel efficiency assumptions
EPA RFS Annual Rule / RVO Setting	EPA	Annual	Usually November	C6-C24 — sets mandatory biofuel blend volumes; affects RIN prices and blending economics

NYMEX RBOB Gasoline occupies a structurally unique position among commodity futures: it is the only major futures contract where a regulatory calendar (the summer/winter specification transition) creates a predictable, structural premium embedded in the curve at a known date each year. This makes the

February-March-April spread complex the most reliably seasonal spread in petroleum markets — a combination of specification-change premium, refinery maintenance supply reduction, and driving season demand build that occurs in the same window annually.

Beyond the specification calendar, RBOB is driven by the physical supply infrastructure (Colonial Pipeline, ARA import arb, Gulf Coast refinery maintenance), the weekly EIA inventory cycle, and the seasonal demand rhythm of the US driving population. The crack spread — RBOB vs WTI — is the unifying measure of refinery economics that determines whether production incentives are expanding or contracting, and it must always be read alongside the inventory trajectory and the specification calendar to interpret whether current spread levels are structurally justified or represent a trading opportunity.